



# Homeownership and voter turnout in u.s. local elections<sup>☆</sup>

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## ABSTRACT

This paper provides evidence that economic self-interest associated with homeownership affects voter turnout in local elections in the United States. Compared to renters, homeowners are financially invested in their communities and are less mobile. Therefore, homeowners should care more about local policies and have incentives to engage actively in local politics. The disparity in political participation between homeowners and renters, however, should diminish in presidential elections for which policy discussions are more targeted at the national-level. These hypotheses are tested using census block group-level election panel data. Fixed effects models and a control function approach are used to identify the effect of homeownership on voter turnout in off-year mayoral elections relative to presidential elections. Results show that mayoral election voter turnout increases with the local homeownership rate. This suggests that local policies may tend to cater to the tastes of homeowners over renters.

## 1. Introduction

Local governments in the United States are the primary provider of local public goods and enact zoning laws that affect allowable patterns of land use. Collectively, local government expenditures also account for one-tenth of U.S. GDP and local governments collect as much in tax revenue as the federal government (Oliver et al., 2012). Nevertheless, compared to national-level elections, U.S. local races often have very low voter turnout. Table 1 summarizes mayoral and presidential election turnout rates from Philadelphia, Seattle and Chicago over the period between 2003 and 2013 (with turnout rates measured at the census block group level).<sup>1</sup> While presidential election turnout averages roughly 60%, mayoral election turnout is much lower, ranging from 20% to 40%. Turnout rate variation across census block groups – measured by national deviation divided by the mean – is also higher in mayoral races. Given local government's consequential role, the limited turnout in local elections is a source of concern. The reason is that voter turnout could affect how local governments enact policies and whether local policies are representative of the electorate (Hajnal and Trounstein, 2005; Hajnal, 2009).

The question of why voter turnout is so low in local elections is related to the literature on the homevoter hypothesis (Fischel, 2001;

Dehring et al., 2008; Brunner et al., 2001; Brunner and Sonstelie, 2003; Hilber and Mayer, 2009; Ahlfeldt, 2011; Ahlfeldt and Maennig, 2015). Previous studies by Fischel (2001) and others have examined how capitalization effects associated with local policies may affect home-voters preferences over local policy initiatives as compared to lease-voters. That literature, however, has largely overlooked related effects on local election turnout. This paper fills that gap by providing evidence that voter turnout in local elections is driven in part by voter economic self-interest related to homeownership status.

For two reasons, homeowners have stronger economic incentives to vote in mayoral elections relative to renters. From an investment perspective, the value of a homeowner's house – the largest investment for most U.S. households<sup>2</sup> – is tied to local fiscal services and amenities provided by the municipal government (Rosen, 1974; Ross and Yinger, 1999; Yinger, 2015). From a consumption perspective, homeowners are also less mobile and hence receive longer utility flows from local public goods. Renters, in contrast, are less financially invested in their communities and more mobile (Rosenthal, 1988; Ioannides and Kan, 1996). Therefore, renters are less likely to internalize the long run effect of their local political decisions and have less incentive to vote in local elections.

This paper empirically test if neighborhoods (census block groups)

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<sup>1</sup> For each city, the statistics are organized into three election cycles and the election cycles are four years apart.

<sup>2</sup> According to a report by The Federal Reserve Board, even at the bottom of 2008 housing crisis, housing wealth still counts as one-half of the total household net wealth in the United States. For the median household, housing wealth counts for almost two-thirds of their total wealth. Link: <https://www.federalreserve.gov/pubs/ifdp/2011/1027/ifdp1027.htm>.

**Table 1**  
Mean and dispersion measure of election turnout rate from three U.S. cities.

| City         | Election cycle | N    | Mayoral election turnout |                      | Presidential election turnout |                      |
|--------------|----------------|------|--------------------------|----------------------|-------------------------------|----------------------|
|              |                |      | $\mu$                    | $\frac{\sigma}{\mu}$ | $\mu$                         | $\frac{\sigma}{\mu}$ |
| Philadelphia | 1              | 1129 | 0.42                     | 0.26                 | 0.62                          | 0.19                 |
|              | 2              | 1129 | 0.25                     | 0.36                 | 0.66                          | 0.24                 |
|              | 3              | 1129 | 0.16                     | 0.43                 | 0.60                          | 0.26                 |
| Chicago      | 1              | 1874 | 0.24                     | 0.45                 | 0.53                          | 0.32                 |
|              | 2              | 1874 | 0.26                     | 0.42                 | 0.61                          | 0.26                 |
|              | 3              | 1874 | 0.33                     | 0.36                 | 0.55                          | 0.32                 |
| Seattle      | 1              | 377  | 0.40                     | 0.35                 | 0.66                          | 0.27                 |
|              | 2              | 377  | 0.45                     | 0.31                 | 0.70                          | 0.23                 |
|              | 3              | 377  | 0.41                     | 0.34                 | 0.69                          | 0.26                 |

*Note:* The election turnout rates are measured at census block level. This election panel covers mayoral and presidential elections held between 2003 and 2013 for Philadelphia, Chicago and Seattle. The elections are grouped into three election cycles and the election cycles are four years apart. Each election cycle contains a mayoral election and a presidential election. The detail of the election grouping is summarized in Table 2.

with higher owner-occupancy rate have higher mayoral election turnout. However, it is well documented in the literature that homeownership and political participation may be endogenous correlated (DiPasquale and Glaeser, 1999; Keyssar, 2009; Engelhardt et al., 2010). Failing to control for unobserved confounders may bias estimates of the effect of the owner-occupancy rate on voter turnout.<sup>3</sup> To address the endogeneity concern, I use two strategies to control for block group level unobservables, a block group-level fixed effects model and a control function approach. Both models use presidential elections as the counterfactual.

National-level elections and related policy initiatives are by definition less focused on local issues, the provision of local public goods, and local property values. For that reason, and drawing on capitalization arguments from the homevoter literature, homeowners and renters should display more similar tendencies to vote in national elections relative to local elections, all else equal. Hence in a well-specified model, block group-level homeownership rates should not be correlated with presidential election turnout provided one sufficiently controls for socioeconomic differences between homeowners and renters. A similar idea is implemented in McCabe (2013). Using individual-level pooled cross sectional data from the Current Population Survey, McCabe (2013) finds that homeowners are more likely than renters to vote in both local and national elections. They also show the difference in tendency to vote is larger in local elections. Differing from McCabe (2013), this paper uses census block group-level panel data to test the hypotheses and controls for confounders using fixed effects models and a control function approach.

In the fixed effects models to follow, I assume that the block group-level confounders are time-invariant. After differencing away time-invariant unobserved confounders, the homeownership rate strongly affects mayoral election turnout but does not affect presidential election turnout. The sharp differences in estimates for mayoral and presidential elections provide evidence that economic incentives contribute to voter

<sup>3</sup> There has been a few attempts in the literature to address the endogeneity. For instance, DiPasquale and Glaeser (1999) and Holian (2011) instruments individual homeownership using group-average homeownership rates from the corresponding socio-demographic groups. However, their instrument is far from perfect since individual unobservables may correlate with membership to a socio-demographic group. Engelhardt et al. (2010) exploits the random assignment of home-purchase subsidies to low-income renters in a field experiment conducted in Tulsa, Oklahoma. Despite a cleaner study design, their sample is not representative of the general population and the sample size is limited.

turnout and motivate homeowners to be more likely to vote in mayoral elections than renters.<sup>4</sup>

One may argue against the assumption that the block group-level confounders are time-invariant. Relaxing this assumption motivates my second identification strategy – the control function (CF) approach. I directly model the time-varying confounders in mayoral regressions as a function of presidential election turnout rate. To obtain identification, the CF approach imposes other moderate assumptions that are clarified later in the paper. Empirically the two models deliver very similar estimates.

An interpretation of my identification strategies is that, by controlling for block group-level unobservables, both models indirectly absorb individual-level confounders that contribute to residential sorting. According to the Tiebout sorting theory, households may sort into neighborhoods according to their needs and willingness to pay for local public amenities (Tiebout, 1956). In a sorting equilibrium, homeowners and renters living in the same neighborhood share lots of similar characteristics. Therefore, census block group-level unobserved attributes may correlate with the individual characteristics that cause the endogenous correlation between homeownership and voter turnout.<sup>5</sup>

To conduct the analysis, I assembled a novel election panel data for Philadelphia, Chicago, and Seattle over the period between 2002 and 2013. My key specification shows that when a census block group switches from fully rental into fully owner-occupied, its mayoral election turnout rate will increase by approximately four percentage points, which is equivalent to a 20 percent increase compared to the mean. The tenure composition change does not affect presidential election turnout.

Results from this paper suggest that renters are under-represented in U.S. local elections. As renters participate less in local races due to insufficient economic self-interest, local politicians may design policies to please the high turnout group – homeowners – to gain electoral support. Such favoritism may lead to policies protecting property value appreciation (e.g. strict zoning laws).<sup>6</sup> Glaeser et al. (2005) points out that change in land regulation regime explains the scarcity of house development in the most expensive U.S. housing market. Ortalo-Magne and Prat (2014) theorizes how homeowners affect urban growth control through the local political process. Total social welfare may also be impaired by the tightening housing supply as it impedes an efficient spatial allocation of labor (Hsieh and Moretti, 2015).

The remainder of the paper is organized as follows. Section 2 gives a detailed discussion of the empirical specializations and identification strategies. Section 3 provides a description of the data source and summary statistics. Section 4 presents the empirical results and robustness check. Then the paper ends with some concluding remarks in Section 5.

## 2. Empirical specification and identification

The basic empirical specification is given as follows:

<sup>4</sup> That interpretation is further strengthened by including household mobility controls (principally census block group residential turnover rates) into the model to help separate investment and consumption motives for voter turnout. Several other time-varying block group level socioeconomic attributes are also taken into account, including education, income, age distribution, share of households that are married, and racial composition. As noted above, the most robust models include block group-level fixed effects and identify off of within-block group temporal variation in the data.

<sup>5</sup> Indeed, Minkoff (2014) finds that the quality of city-provided public goods in a community is highly correlated with residents tendency to vote in New York City.

<sup>6</sup> In San Francisco, a city with a roughly thirty-five percent owner-occupancy rate, households organize into hundreds of politically powerful neighborhood groups (e.g., Telegraph Hill Dwellers) to promote policies limiting new house development. See Kim-Mai Cutler, “How Burrowing Owls Lead To Vomiting Anarchists (Or SF’s Housing Crisis Explained)”, TechCrunch, April 2014. Link: <http://techcrunch.com/2014/04/14/sf-housing/>.

**Table 2**  
Election panel grouping list and merge with Census data.

| Election cycle | Mayoral election                     | Presidential election | Census data                                                            |
|----------------|--------------------------------------|-----------------------|------------------------------------------------------------------------|
| 1              | 2003 Philly, Chicago                 | 2004 All three cities | Average of 2000 Decennial Census<br>and ACS 2005–2009 5-year-estimates |
|                | 2005 Seattle<br>2007 Philly, Chicago |                       |                                                                        |
| 2              | 2009 Seattle<br>2011 Philly, Chicago | 2008 All three cities | ACS 2006–2010 5-year-estimates                                         |
|                | 2013 Seattle                         |                       |                                                                        |
| 3              |                                      | 2012 All three cities | ACS 2010–2014 5-year-estimates                                         |

*Note:* The election panel covers mayoral and presidential elections held between 2003 and 2013 for Philadelphia (Philly), Chicago and Seattle. Data merge between election data and Census data is based on 2010 Census blocks definition.

$$\text{Mayoral Election turnout}_{b,c,t} = \alpha_m + \beta_m \times \text{Owner-occupancy rate}_{b,c,t} + \mathbf{X}'_{b,c,t} \boldsymbol{\Gamma}_m + u_{b,c,t} \quad (1)$$

in which  $b$  indexes census block groups,  $c$  cities and  $t$  election cycles. Coefficient  $\beta_m$  measures the effect of owner-occupancy rate on mayoral election voter turnout. This paper attempts to obtain an unbiased estimator of  $\beta_m$ . Vector  $\mathbf{X}_{b,c,t}$  represents observed census block group level attributes that may correlate with owner-occupancy rate and mayoral election turnout. The controls to be included in  $\mathbf{X}_{b,c,t}$  will be introduced in Section 4. After controlling for the observables, standard OLS theory shows that the estimator of  $\beta_m$  is unbiased if owner-occupancy rate and the remaining error term  $u_{b,c,t}$  are independent. Nevertheless,  $u_{b,c,t}$  may still contain unobserved confounders that are correlated with owner-occupancy rate. I further decompose  $u_{b,c,t} = \gamma_m \delta_{b,c,t} + \varepsilon_{b,c,t}$  and define  $\delta_{b,c,t}$  as the unobserved block group-level confounders. Coefficient  $\gamma_m$  measures the correlation between the confounders and mayoral election turnout. Since  $\delta_{b,c,t}$  is unobserved to the researcher, this paper uses two strategies to control for it.

### 2.1. Fixed effects model

The first approach exploits the panel structure of the data. Assuming that the unobserved confounders are time-invariant, I use a fixed effects model to absorb the unobservables. Replacing  $\delta_{b,c,t}$  by  $\delta_{b,c}$ , Eq. (1) can be written as:

$$\text{Mayoral Election turnout}_{b,c,t} = \alpha_m + \beta_m \times \text{Owner-occupancy rate}_{b,c,t} + \mathbf{X}'_{b,c,t} \boldsymbol{\Gamma}_m + \gamma_m \delta_{b,c} + \pi_t + \varepsilon_{b,c,t} \quad (2)$$

in which I also include election-cycle fixed effects  $\pi_t$  to capture time trend in the data. After the unobservables been absorbed by block group-level fixed effects, an unbiased estimator of  $\beta_m$  can be obtained.

The identification assumption in the fixed effects model is that the unobserved confounders are time-invariant. To support this assumption, I use presidential election outcome as the counterfactual. Since national-level politics are less relevant to property value and local public good provision, the disparate incentive to vote between homeowners and renters should be reduced. Therefore, if the fixed effects can absorb block group-level unobserved confounders, homeownership rate should not drive presidential election turnout in the fixed effects model. Replacing the dependent variable in Eq. (2) with presidential election turnout, I have:

$$\text{Presidential Election turnout}_{b,c,t} = \alpha_p + \beta_p \times \text{Owner-occupancy rate}_{b,c,t} + \mathbf{X}'_{b,c,t} \boldsymbol{\Gamma}_p + \gamma_p \delta_{b,c} + \pi_t + \varepsilon_{b,c,t} \quad (3)$$

In Eq. (3), the same time-invariant confounder term  $\delta_{b,c}$  may correlate with owner-occupancy rate and thus generates an endogenous correlation between owner-occupancy rate and presidential election

turnout. Coefficient  $\gamma_p$  captures the effect of unobservables on presidential election turnout and it is allowed to be different from  $\gamma_m$  in Eq. (2). Coefficient  $\beta_p$  measures the impact of owner-occupancy rate on presidential election turnout. As will be shown in the empirical section,  $\beta_p$  becomes statistically indistinguishable from zero once the fixed effects are introduced.

### 2.2. Control function approach

One may argue against the assumption that the block group-level confounders are time-invariant. Relaxing this assumption motivates my second identification strategy – the control function (CF) approach. The basic idea of a CF approach is to control for the unobservables in the error term by modeling them directly (Wooldridge, 2015). Since I can observe turnout rates from both mayoral and presidential elections, I can re-write the confounder term  $\delta_{b,c,t}$  in mayoral election regression as a function of presidential election turnout rate. To better explain the CF approach, I rewrite Eqs. (2) and (3) as:

$$\text{Mayoral election turnout}_{b,c,t} = \alpha_m + \beta_m \times \text{Owner-occupancy rate}_{b,c,t} + \gamma_m \delta_{b,c,t} + \varepsilon_{b,c,t} \quad (4)$$

and

$$\text{Presidential election turnout}_{b,c,t} = \alpha_p + \beta_p \times \text{Owner-occupancy rate}_{b,c,t} + \gamma_p \delta_{b,c,t} + \varepsilon_{b,c,t} \quad (5)$$

Vector  $\mathbf{X}_{b,c,t}$  is dropped for simplicity. The confounder term  $\delta_{b,c,t}$  enters both equations and is allowed to have heterogeneous impact across elections (measured by  $\gamma_m$  and  $\gamma_p$ , respectively). My coefficient of interest is  $\beta_m$ , which measures the impact of owner-occupancy rate on mayoral election turnout. Notice that neither  $\gamma_m$  or  $\gamma_p$  can be estimated without bias because  $\delta_{b,c,t}$  is unobserved. Rearranging Eq. (5), I can write  $\delta_{b,c,t}$  as a function of the *Presidential election turnout* <sub>$b,c,t$</sub> . Plugging the new expression of  $\delta_{b,c,t}$  into Eq. (6), I arrive at:

$$\text{Mayoral election turnout}_{b,c,t} = A + \left( \beta_m - \frac{\gamma_m}{\gamma_p} \beta_p \right) \times \text{Owner-occupancy rate}_{b,c,t} + \frac{\gamma_m}{\gamma_p} \text{Presidential turnout}_{b,c,t} + v_{b,c,t} \quad (6)$$

in which  $A = \alpha_m - \frac{\alpha_p}{\gamma_p}$  and  $v_{b,c,t} = \varepsilon_{b,c,t} - \frac{\gamma_m}{\gamma_p} \varepsilon_{b,c,t}$ . Now all the controls in Eq. (6) can be observed from the data. More important, the owner-occupancy rate is independent from the new error term. Therefore, I can obtain an unbiased estimator of the coefficient  $\beta_m - \frac{\gamma_m}{\gamma_p} \beta_p$ . This is the difference between my coefficient of interest  $\beta_m$  and a scaled version of  $\beta_p$ . Before introducing the identification assumptions, let me briefly discuss the sign of the scale parameter  $\frac{\gamma_m}{\gamma_p}$ . Notice that although  $\frac{\gamma_m}{\gamma_p}$  also appears as the coefficient of presidential election turnout rate, it cannot be estimated without bias because the presidential election turnout is

correlated with the error term by construction. However, it is reasonable to presume that the sign of  $\frac{\gamma_m}{\gamma_p}$  is positive since  $\gamma_m$  and  $\gamma_p$  measure the endogenous correlation between owner-occupancy rate and election turnout rates. In the empirical section, I find that the estimate of  $\frac{\gamma_m}{\gamma_p}$  is approximately 0.45 and it remains stable in all specifications.

To identify  $\beta_m$ , I need to make one of the two alternative assumptions. One option is assuming  $\beta_p = 0$ . In this case  $\beta_m - \frac{\gamma_m}{\gamma_p}\beta_p = \beta_m$ , so that an OLS regression based on Eq. (6) returns an unbiased estimate of  $\beta_m$ . The other option is assuming  $\beta_p$  to be non-negative. Under this weaker assumption and the fact that  $\frac{\gamma_m}{\gamma_p}$  is positive as discussed above, I can identify a lower bound of  $\beta_m$ . Both assumptions imply that conditional on  $\delta_b, c, t$  the owner-occupancy rate has a negligible effect on presidential election turnout. Neither of these assumptions is restrictive. According to the fixed effects model estimation results to be shown in Section 4.1, the correlation between owner-occupancy rate and presidential election turnout is not statistically different from zero. Since CF approach further allows  $\delta_b, c, t$  to be time-variant, this more flexible specification should further push  $\beta_p$  to be close to zero. Therefore, depending on which assumption one is willing to make, either  $\beta_m$  or its lower bound is identified.<sup>7</sup> For the ease of discussion, I assume that  $\beta_p = 0$ , therefore  $\beta_m$  can be identified from an OLS regression using Eq. (6).

### 3. Data source and summary statistics

This paper draws upon data from the U.S. Census and election databases from Philadelphia, Chicago, and Seattle.<sup>8</sup> Municipal governments from these three cities digitized their precinct-level election returns from 2002 onward and published them online (see Appendix B for details). The election database reports election type, candidate information, and vote counts for local, state, and federal elections. It also includes records from various idiosyncratic local ballots.

Now I discuss the reasons to choose mayoral and presidential elections as the focus of this study. First, my main identification strategy relies upon using fixed effects to absorb block group-level time-invariant confounders. Hence I need to look at elections that were held for multiple times. Second, choosing elections that are common for all three cities allows me to pool data together to improve estimation precision. Presidential elections are a natural candidate for national elections since they are held every four years and are universal for all residents living in the United States. For local elections, I chose mayoral elections because all three cities have the same mayor-council (as opposed to council-manager) governance system that allows city residents to elect their mayors directly. Additionally, all three cities hold their mayoral elections in odd-numbered years (off-year elections) so that the mayoral election turnout is not affected by presidential election turnout (Levonyan, 2013). Another potential candidate for local election is the city-council election. In my data, the city-council and the mayoral election turnout rates are close because they are held at the same time. Voters, given that the marginal cost of participating in city-council elections is zero conditional on already voted in a mayoral race, usually cast votes for both elections. Since mayor tends to be a more viable political player in the mayor-council system (Holbrook and Weinschenk, 2013) and it is the local executive parallel to the U.S. chief executive, this paper focuses on mayoral election results. Other elections such as gubernatorial or congressional elections are not as relevant for local policy as mayoral elections and are therefore not considered in the analysis.

<sup>7</sup> Identification fails only when owner-occupancy rate has a notable negative impact on presidential election turnout. In that case the estimated coefficient is an upper bound of  $\beta_m$ . However, based on the empirical results in Section 4.1 I can rule out this scenario.

<sup>8</sup> These are large central city samples where the homevoter effect may be the weakest as compared to suburbs or rural areas (Schleicher, 2010). So that the estimates obtained in this paper may be a lower bound of the homevoter effect.

I group the election turnout data into three election cycles according to chronological order. Each election cycle contains a mayoral election and a presidential election that is one year apart. Between 2002 and 2015, Philadelphia and Chicago held mayoral elections in 2003, 2007 and 2011; Seattle held mayoral elections in 2005, 2009 and 2013. Correspondingly, there are three presidential elections in 2004, 2008 and 2012. The first three columns in Table 2 summarizes the election grouping. For instance, the first election cycle includes 2003 Philadelphia, 2003 Chicago, and 2005 Seattle mayoral elections as well as the 2004 presidential election from these three cities.

To merge the election data with the Census data, I map the raw precinct-level voting records into census block groups (2010 definition) using a crosswalk from the Missouri Census Data Center.<sup>9</sup> Census blocks are the smallest geographic unit for which the Census publishes its data. A census block group typically has a population between 600 and 3,000. The average census block group population size is 1000 for the three cities. Voting precincts are slightly smaller than census block groups geographically. In Philadelphia, 1686 precincts are mapped into 1333 block groups. In Chicago, 2571 precincts are mapped into 2171 block groups and, in Seattle, 960 precincts are mapped into 479 block groups.

Census data at the block group level is accessible from two Census data products: the Decennial Census and the American Community Survey (ACS) 5-year-estimates. The main difference between them is that the Decennial Census provides data gathered at a “point of time” while ACS produces estimates using data gathered within a “period of time”. Since socio-demographic attributes of small geographic areas are likely to remain constant within a 5-year time window, ACS 5-year-estimates are designed to provide estimates describing the average attributes of an area over the corresponding time period.<sup>10</sup> Moreover, Decennial Census data is available once per decade while ACS 5-year-estimates have more frequent time coverage.<sup>11</sup> The data merging between election panel and Census data is summarized in Table 2 Column (3). There is no corresponding Census data coverage for elections held during the first election cycle (2003 to 2005). Hence I approximate the block group-level attributes during this period by taking the average between 2000 Decennial Census data and ACS 2005–2009 5-year-estimates.<sup>12</sup> For election data from the second and the third election cycles, I merge them with Census data from ACS 2006–2010 5-year-estimates and ACS 2010–2014 5-year-estimates, respectively.<sup>13</sup>

The block group-level voter turnout rate is measured by the total number of votes over the size of voting age population (United States citizens 18 years of age or older) in the block group.<sup>14</sup> Due to sampling

<sup>9</sup> Link: <http://mcde.missouri.edu/websas/geocorr12.html>.

<sup>10</sup> For more information of ACS data, please look at the “A Compass for Understanding and Using American Community Survey Data” document prepared by Census. Link: [http://www.psc.isr.umich.edu/dis/acs/handouts/Compass\\_Appendix.pdf](http://www.psc.isr.umich.edu/dis/acs/handouts/Compass_Appendix.pdf).

<sup>11</sup> Currently ACS has 6 rounds of 5-year-estimates available: 2005–2009, 2006–2010, 2007–2011, 2008–2012, 2009–2013, 2010–2014.

<sup>12</sup> The approximation can be done for all variables except the neighborhood turnover rate, which measures share of residents moved into the neighborhood within past 12 months. It is available in ACS data but not in 2000 Decennial Census, which only provides the share of residents moved into the neighborhood within past five years. Neighborhood turnover rate appears to have a slightly downward trend over the years. It is mean value decreases from 0.165 in 2005–2009 round data to 0.158 in 2010–2014 round data. One option for obtaining approximation is extrapolating the original sequence by assuming linearity. Since the estimation will include election cycle fixed effects which can take care any form of linear/non-linear trend in variables, this paper directly uses 2005–2009 data to proximate the neighborhood turnover rate in first election cycle.

<sup>13</sup> All the data merging is based on the 2010 geographic definitions of census block groups. However, 2000 Decennial Census and ACS 2005–2009 5-year-estimates files are coded using 2000 census block groups boundaries while other years' ACS data are coded using 2010 census block groups boundaries. The Census Bureau does not provide correspondence file to link the 2000 census block groups to the 2010 census block groups. Therefore, I build correspondence file between 2000 and 2010 census block groups using GIS software (see Appendix C for detailed GIS work description).

<sup>14</sup> Voting age population estimates are obtained from a separate tabulation from 2000 Decennial Census and the “Voting Age Population by Citizenship and Race (CVAP)” estimates from the ACS 5-year-estimates. See Appendix B for details.



**Table 3**  
Summary statistics for election turnout and owner-occupancy rate.

| City    | Election cycle | N    | Mayoral election turnout mean/median (sd) | Presidential election turnout mean/median (sd) | Owner-occupancy rate mean/median (sd) |
|---------|----------------|------|-------------------------------------------|------------------------------------------------|---------------------------------------|
| Philly  | 1              | 1129 | 0.42/0.43 (0.11)                          | 0.62/0.63 (0.12)                               | 0.59/0.61 (0.21)                      |
|         | 2              | 1129 | 0.25/0.25 (0.09)                          | 0.66/0.65 (0.16)                               | 0.57/0.58 (0.23)                      |
|         | 3              | 1129 | 0.16/0.16 (0.07)                          | 0.60/0.59 (0.16)                               | 0.54/0.55 (0.23)                      |
| Chicago | 1              | 1874 | 0.24/0.23 (0.11)                          | 0.53/0.53 (0.17)                               | 0.48/0.45 (0.23)                      |
|         | 2              | 1874 | 0.26/0.24 (0.11)                          | 0.61/0.61 (0.16)                               | 0.49/0.47 (0.24)                      |
|         | 3              | 1874 | 0.33/0.31 (0.12)                          | 0.55/0.53 (0.18)                               | 0.47/0.44 (0.24)                      |
| Seattle | 1              | 377  | 0.40/0.41 (0.14)                          | 0.66/0.69 (0.18)                               | 0.49/0.50 (0.25)                      |
|         | 2              | 377  | 0.45/0.46 (0.14)                          | 0.70/0.71 (0.17)                               | 0.49/0.50 (0.26)                      |
|         | 3              | 377  | 0.41/0.43 (0.14)                          | 0.69/0.72 (0.18)                               | 0.49/0.50 (0.25)                      |

*Note:* The unit of observation is census blocks. The election panel covers mayoral and presidential elections held between 2003 and 2013 for Philadelphia (Philly), Chicago and Seattle. The owner-occupancy rate data is obtained from the Census.

error in the voting age population estimation, there are some block groups with turnout rates larger than one in some years. In the empirical analysis to follow, I only use block group samples in which the turnout rate is never larger than one. For instance, if block group A has a turnout rate larger than one in the presidential election from the third election cycle, all records from block group A are dropped. After dropping those block groups I obtain a balanced election panel in which each election cycle is composed of 1129 Philadelphia census block groups, 1874 Chicago census block groups and 377 Seattle census block groups. In the robustness checks section, I show that all results hold using the original full sample, as well as to as when restricting the sample to block groups with turnout rates between 5% and 95%.

Table 3 provides mean, median and standard deviation for homeownership rate and turnout in mayoral and presidential elections. The first thing to notice is that in each of the cells the mean and median are close to each other, indicating that the turnout rates do not have a skewed distribution. Compared to presidential election turnout, mayoral election turnout is lower.

## 4. Empirical results

### 4.1. Fixed effects model estimation results

The empirical evidence from the fixed effects model is presented in three parts. First, I present estimation results from Eqs. (2) and (3). This allows me to compare the effect of owner-occupancy rate on mayoral elections ( $\hat{\beta}_m$ ) versus on presidential elections ( $\hat{\beta}_p$ ). Second, I statistically test the coefficient difference between  $\hat{\beta}_m$  and  $\hat{\beta}_p$  by pooling mayoral and presidential elections data together to run regressions with interaction terms. Last, I present further evidence of obtaining identification using the fixed effects model.

Table 4 presents estimation results based on Eq. (2) which uses mayoral election turnout rate as the dependent variable. Robust standard errors are clustered at the census block group level to account for time series correlation. The first column shows the raw correlation between owner-occupancy rate and mayoral election turnout. City fixed effects are added to account for city-level heterogeneity. The coefficient on the owner-occupancy rate is positive and significant. It indicates that when a block group switches from fully rental into fully owner-occupied, its mayoral election turnout rate increases by 19.7% points, which is almost a 70% increase relative to the mean.

Column (2) adds in the mobility control. As described in the introduction, owning a property makes homeowners have a higher financial stake in the community as well as be less mobile. Staying put means homeowners enjoy a longer utility flow from local public good. This may also motivate them to vote in local elections to affect public goods provision.<sup>15</sup> Controlling for the residential mobility measure

aims to separate the effect of investment and consumption motive on voter turnout. However, there is no direct measure of residential mobility at block group-level. I use census block group turnover rate - the share of residents moving into the neighborhood within past 12 months - as a proxy for the mobility measure, by assuming that people living in high turnover neighborhoods are in general more mobile. Being aware that the turnover rate is not an adequate mobility measure, I also rely on other covariates to be introduced later to control for residential mobility.<sup>16</sup> Not surprisingly, controlling for neighborhood turnover rate drives down the owner-occupancy rate coefficient. The coefficient on owner-occupancy rate declines from 0.197 to 0.158. The negative coefficient on the mobility control is consistent with the idea that mobile residents have less incentive to participate in local elections.

Column (3) substitutes city fixed effects with census block group fixed effects. Fixed effects absorb time-invariant block group-level unobserved confounders that may cause an endogenous correlation between homeownership rate and voter turnout. In the block group fixed effects model, coefficient estimates on owner-occupancy rate and mobility control both decrease.

From Column (4) to Column (8), I gradually include controls for block group-level observables into the specification. The first set of controls to be added is education distribution. Homeownership is highly correlated with education according to a report from First American Financial Corporation.<sup>17</sup> Numerous studies also find that education is a strong driver of election turnout (Sondheimer and Green, 2010; Burden, 2009). Looking at residents 25 years old or above, I generate the share of residents having a high school degree or some college and the share of residents with at least a bachelor's degree. These two variables jointly measure the education distribution of the census block group. Results from Column (4) show that there is a non-monotonic relationship between education attainment and voter turnout. Compared to residents with less than a high school degree, the share of residents with high school or some college is negatively correlated with turnout while the share of residents with at least a college degree positively correlates with voter turnout. The effect of homeownership on mayoral election turnout slightly decreases.

Column (5) adds in for the household median income. According to an analysis done by the real estate database company Zillow<sup>18</sup>, owner-occupancy rate almost monotonically increases with household income. Meanwhile, higher income may have a negative impact on voter turnout. Charles and Stephens (2013) shows that higher labor income discourages voter turnout because higher hourly wages implies higher

(footnote continued)

for most states is 15–30 days before the Election day, as long as the majority of the movers do not frequently move immediately before the Election day, the effect of mobility on voter registration should be small.

<sup>16</sup> In a comprehensive review, Molloy et al. (2011) shows that U.S. residential mobility is closely tied to demographic attributes such as education or age.

<sup>17</sup> Link: <http://www.firstam.com/economics/homeownership-progress-index/>.

<sup>18</sup> Link: <http://www.zillow.com/research/homeownership-by-income-9419/>.

<sup>15</sup> One may also suspect that moving too frequently may bring hurdle to voter registration and thus depress election turnout. Given the fact that voter registration deadline

**Table 4**  
Fixed effects model estimation results using mayoral election panel data.

|                                                     | (1)                | (2)                 | (3)                | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.197***<br>(0.01) | 0.158***<br>(0.01)  | 0.078***<br>(0.01) | 0.064***<br>(0.01)  | 0.097***<br>(0.02)  | 0.050***<br>(0.02)  | 0.047***<br>(0.02)  | 0.038**<br>(0.02)   |
| % Moved into current residence within past 12 month |                    | −0.167***<br>(0.01) | −0.048**<br>(0.02) | −0.054***<br>(0.02) | −0.056***<br>(0.02) | −0.012<br>(0.02)    | −0.012<br>(0.02)    | −0.017<br>(0.02)    |
| % High school and some college among pop > 25       |                    |                     |                    | −0.119***<br>(0.02) | −0.110***<br>(0.02) | −0.090***<br>(0.02) | −0.088***<br>(0.02) | −0.104***<br>(0.02) |
| % College above among pop > 25                      |                    |                     |                    | 0.004<br>(0.02)     | 0.042*<br>(0.02)    | 0.082***<br>(0.02)  | 0.083***<br>(0.02)  | 0.040<br>(0.03)     |
| Household median income (Million \$)                |                    |                     |                    |                     | −0.765***<br>(0.12) | −0.735***<br>(0.12) | −0.767***<br>(0.12) | −0.794***<br>(0.12) |
| % Adults 30 to 60                                   |                    |                     |                    |                     |                     | 0.211***<br>(0.02)  | 0.209***<br>(0.02)  | 0.209***<br>(0.02)  |
| % Adults 60 above                                   |                    |                     |                    |                     |                     | 0.302***<br>(0.03)  | 0.301***<br>(0.03)  | 0.279***<br>(0.03)  |
| % Married-household                                 |                    |                     |                    |                     |                     |                     | 0.020<br>(0.02)     | 0.020<br>(0.02)     |
| % Hispanic                                          |                    |                     |                    |                     |                     |                     |                     | −0.006<br>(0.06)    |
| % White                                             |                    |                     |                    |                     |                     |                     |                     | 0.072<br>(0.06)     |
| % Black                                             |                    |                     |                    |                     |                     |                     |                     | −0.044<br>(0.06)    |
| % Asian                                             |                    |                     |                    |                     |                     |                     |                     | −0.028<br>(0.06)    |
| R-squared                                           | 0.24               | 0.26                | 0.06               | 0.07                | 0.08                | 0.10                | 0.10                | 0.11                |
| Election cycle fixed effects                        | 3                  | 3                   | 3                  | 3                   | 3                   | 3                   | 3                   | 3                   |
| City fixed effects                                  | 3                  | 3                   | —                  | —                   | —                   | —                   | —                   | —                   |
| Census block group fixed effects                    | —                  | —                   | 3369               | 3369                | 3369                | 3369                | 3369                | 3369                |
| Observations                                        | 10107              | 10107               | 10107              | 10107               | 10107               | 10107               | 10107               | 10107               |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

opportunity cost of going to the poll. The coefficient estimate suggests that block groups with higher median household income have lower turnout rate, consistent with [Charles and Stephens \(2013\)](#)'s finding. The coefficient estimate on owner-occupancy rate increases.

Column (6) further controls for the age distribution. Age is another strong predictor of homeownership and it is well documented in the voting literature that propensity to vote increases with age ([Blais, 2000](#); [Wolfinger and Rosenstone, 1980](#)). I use the share of adults between 30 and 60 years old and the share of adults above 60 years old to jointly measure the age distribution of the census block group. The age distribution coefficients indicate that voter turnout is positively correlated with voter age. Adults above 60 years old have strongest tendency to vote among all groups. The coefficient on owner-occupancy rate drops from 9.7% to 5%.

Column (7) further contains the share of married household. It has been found that marriage is closely linked to the first-time transition into homeownership ([Smits and Mulder, 2008](#)). Marriage may affect turnout through a complex web of channels. For instance, marriage may depress turnout if married couples need to spend time with their children at home, which prevents them from going to the polls. On the other hand, if couples have school-aged children that they drop off to school and the polling stations are close to local schools, it might be convenient for them to vote. In this case, marriage may increase voter turnout.<sup>19</sup> The share of married household does not seem to have a significant correlation with mayoral election turnout. The effect of owner-occupancy rate on mayoral

election turnout remain stable.

Finally, Column (8) brings the last set of census block group controls – racial composition. This aims to alleviate concerns that the home-ownership effect may be caused by systematic differences in turnout across racial groups. None of the coefficients on racial composition controls are significant, and the coefficient estimate on owner-occupancy rate remain largely unchanged. With the full set of controls, owner-occupancy rate still has a notable impact on mayoral election turnout. Holding all else equal, if a block group switches from fully rental into fully owner-occupied, its mayoral turnout rate increases by 3.8% points. This is equivalent to a 18 percent increase compared to the mean.<sup>20</sup>

Having established that owner-occupancy rate drives mayoral elections turnout in a fixed effects model, I move to the benchmark presidential election results in [Table 5](#). The first column of [Table 5](#) presents the raw correlation between owner-occupancy rate and presidential election turnout rate. According to the coefficient estimate, when a block group switches from fully rental into fully owner-occupied, its presidential election turnout increases by 9.6% points. This is a 17.5% increase compared to the average presidential turnout rate. Adding in mobility controls in Column (2) does not change the owner-occupancy rate coefficient by much. In Column (3), replacing city fixed effects with census block group fixed effects erases the previously significant coefficient on owner-occupancy rate. Block group fixed effects capture the endogenous correlation between homeownership and presidential election turnout. The fixed effects also reduce the coefficient

<sup>19</sup> One may argue that it is the existence of school-aged children that matters for married households, since families with children may care more about local public goods provision (such as police, sidewalks, school quality, and so forth.) and therefore more likely to vote. Empirically I find that controlling the share of households with school-aged children instead of the share of married household does not affect the result.

<sup>20</sup> I also conduct the similar analysis using the log of mayoral election turnout rate as the dependent variable to account for potential non-linearity. The results are very similar to the linear model.

**Table 5**  
Fixed effects model estimation results using presidential election panel data.

|                                                     | (1)                | (2)                 | (3)              | (4)                | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|------------------|--------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.096***<br>(0.01) | 0.070***<br>(0.01)  | −0.016<br>(0.02) | −0.027<br>(0.02)   | 0.016<br>(0.02)     | −0.038*<br>(0.02)   | −0.026<br>(0.02)    | −0.034*<br>(0.02)   |
| % Moved into current residence within past 12 month |                    | −0.108***<br>(0.02) | −0.035<br>(0.02) | −0.039*<br>(0.02)  | −0.042*<br>(0.02)   | 0.009<br>(0.02)     | 0.008<br>(0.02)     | 0.004<br>(0.02)     |
| % High school and some college among pop > 25       |                    |                     |                  | −0.053**<br>(0.03) | −0.041<br>(0.03)    | −0.019<br>(0.03)    | −0.025<br>(0.03)    | −0.031<br>(0.03)    |
| % College above among pop > 25                      |                    |                     |                  | 0.030<br>(0.03)    | 0.079***<br>(0.03)  | 0.124***<br>(0.03)  | 0.120***<br>(0.03)  | 0.092***<br>(0.03)  |
| Household median income (Million \$)                |                    |                     |                  |                    | −0.987***<br>(0.14) | −0.962***<br>(0.14) | −0.857***<br>(0.15) | −0.856***<br>(0.15) |
| % Adults 30 to 60                                   |                    |                     |                  |                    |                     | 0.253***<br>(0.03)  | 0.261***<br>(0.03)  | 0.266***<br>(0.03)  |
| % Adults 60 above                                   |                    |                     |                  |                    |                     | 0.342***<br>(0.03)  | 0.345***<br>(0.03)  | 0.335***<br>(0.03)  |
| % Married-household                                 |                    |                     |                  |                    |                     |                     | −0.065***<br>(0.02) | −0.071***<br>(0.02) |
| % Hispanic                                          |                    |                     |                  |                    |                     |                     |                     | 0.059<br>(0.07)     |
| % White                                             |                    |                     |                  |                    |                     |                     |                     | 0.094<br>(0.07)     |
| % Black                                             |                    |                     |                  |                    |                     |                     |                     | −0.023<br>(0.07)    |
| % Asian                                             |                    |                     |                  |                    |                     |                     |                     | 0.070<br>(0.07)     |
| R-squared                                           | 0.10               | 0.10                | 0.07             | 0.07               | 0.08                | 0.10                | 0.11                | 0.11                |
| Election cycle fixed effects                        | 3                  | 3                   | 3                | 3                  | 3                   | 3                   | 3                   | 3                   |
| City fixed effects                                  | 3                  | 3                   | —                | —                  | —                   | —                   | —                   | —                   |
| Census block group fixed effects                    | —                  | —                   | 3369             | 3369               | 3369                | 3369                | 3369                | 3369                |
| Observations                                        | 10107              | 10107               | 10107            | 10107              | 10107               | 10107               | 10107               | 10107               |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

on the mobility control. People's tendency to vote in the presidential election should not be affected by how mobile they are since national policies affect the well-being of all residents in the country. Including the full set of control in Column (4)–(8) does not result in any significant changes in the owner-occupancy rate coefficient.

Column (8) from Table 4 and Column (8) from Table 5 offer a sharp comparison. After addressing the endogeneity concern using fixed effects, owner-occupancy rate is a strong driver of mayoral election turnout but not presidential election turnout. These results confirm the hypothesis that, compared to renters, homeowners are more likely to vote in local elections due to economic incentives.

Then I pool mayoral and presidential elections data together to statistically test the coefficient difference between  $\hat{\beta}_m$  and  $\hat{\beta}_p$ . I define a mayoral election dummy – equaling 1 for data from mayoral elections and 0 for data from presidential elections. I regress voter turnout on the mayoral dummy, the owner-occupancy rate, the interaction term between the two, other controls, fixed effects, and interactions between the mayoral dummy and each of the other aforementioned controls. Table 6 summarizes the pooled regression results. The first row reports the coefficient on the interaction term between the mayoral dummy and the owner-occupancy rate. It measures the difference between  $\hat{\beta}_m$  and  $\hat{\beta}_p$ . A positive estimate indicates that  $\hat{\beta}_m$  is larger than  $\hat{\beta}_p$ . In all specifications, the interaction coefficient estimates remain positive and significant. This implies owner-occupancy rate is a strong driver of mayoral election turnout as opposed to presidential election turnout. The second row reports the coefficient on the mayoral dummy. Unsurprisingly, the coefficient is negative in all columns since mayoral elections have lower turnout rates. The third row presents the effect of owner-occupancy rate on presidential turnout. The coefficient remains

statistically indistinguishable from zero in most columns. It becomes negative in Columns (2), but the effect goes away with a full set of controls in Columns (3). Results from Table 6 confirm the previous findings.

Last, I provide further evidence to support the identification assumptions in the fixed effects model. As discussed in the introduction, one interpretation of controlling for block group-level unobserved attributes is that it indirectly absorbs personal confounders that contribute to residential sorting. If individuals sort into narrowly defined geographic areas, a coarsely defined geographic fixed effect should not be able to correct the endogeneity bias. This implies that, by substituting for geographically refined fixed effects with geographically broader fixed effects, the endogenous correlation between owner-occupancy rate and voter turnout should emerge. In this paper, the most refined geographic level is a census block group. The mean size of census block groups in the data is around 1000 residents, approximately the size of a U.S. city neighborhood (roughly two to three street block groups in densely developed urban areas). A natural choice for a geographic unit that is coarser than a census block group yet more refined than a city is the census tract. A census tract is usually composed of three to four census block groups. At the tract level, uneven distribution of public goods and amenities also exists. Massey (2001) demonstrates that the tract-level “dissimilarity index”<sup>21</sup> is relatively high in both Philadelphia and Chicago (Seattle is not covered in their study). The census tract fixed effects are therefore well suited for an

<sup>21</sup> “Dissimilarity index” measures the relative number of Blacks who would have to change geographic units so that an even Black-White spatial distribution could be achieved.

**Table 6**

Fixed effects model results using pooled mayoral and presidential election panel data.

|                                                            | (1)                 | (2)                 | (3)                 |
|------------------------------------------------------------|---------------------|---------------------|---------------------|
| Owner_occupancy rate*Mayoral dummy                         | 0.078***<br>(0.01)  | 0.078***<br>(0.01)  | 0.072***<br>(0.02)  |
| Mayoral election dummy                                     | –0.296***<br>(0.00) | –0.296***<br>(0.00) | –0.171***<br>(0.03) |
| Owner-occupancy rate                                       | –0.000<br>(0.01)    | –0.037**<br>(0.02)  | –0.034<br>(0.02)    |
| % Moved into current residence within past 12 month        |                     | –0.007<br>(0.02)    | 0.004<br>(0.03)     |
| % High school and some college among pop > 25              |                     | –0.067***<br>(0.02) | –0.031<br>(0.03)    |
| % College above among pop > 25                             |                     | 0.066***<br>(0.03)  | 0.092***<br>(0.04)  |
| Household median income (Million \$)                       |                     | –0.825***<br>(0.12) | –0.859***<br>(0.16) |
| % Adults 30 to 60                                          |                     | 0.238***<br>(0.02)  | 0.266***<br>(0.03)  |
| % Adults 60 above                                          |                     | 0.307***<br>(0.03)  | 0.336***<br>(0.04)  |
| % Married-household                                        |                     | –0.026<br>(0.02)    | –0.071***<br>(0.03) |
| % Hispanic                                                 |                     | 0.026<br>(0.06)     | 0.020<br>(0.07)     |
| % White                                                    |                     | 0.083<br>(0.06)     | 0.054<br>(0.07)     |
| % Black                                                    |                     | –0.034<br>(0.06)    | –0.062<br>(0.07)    |
| % Asian                                                    |                     | 0.021<br>(0.06)     | 0.021<br>(0.06)     |
| R-squared                                                  | 0.70                | 0.71                | 0.77                |
| Election cycle fixed effect                                | 3                   | 3                   | 3                   |
| Census block group fixed effect                            | 3369                | 3369                | 3369                |
| Mayoral dummy interaction with election cycle fixed effect | Yes                 | Yes                 | Yes                 |
| Mayoral dummy interaction with other slope variables       | –                   | –                   | Yes                 |
| Mayoral dummy interaction with block group fixed effect    | –                   | –                   | Yes                 |
| Observations                                               | 20214               | 20214               | 20214               |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census blocks in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

alternative specification using a broader geographic measure. Beyond census tracts, I use city fixed effects as the coarsest geographic unit control.

Table 7 summarizes the findings with different levels of fixed effects. Panel A uses only the mayoral election data while Panel B uses only the presidential election data. The full set of block group-level controls is included in all specifications. Only the coefficient estimates on owner-occupancy rate are reported. Starting from Panel A, Column (1) presents the raw correlation between owner-occupancy rate and mayoral election turnout rate conditional on block group attributes. Adding election cycle fixed effects in Column (2) brings limited change. Column (3) further adds city fixed effects and the coefficient largely remains stable. Column (4) substitutes city fixed effects with census tract fixed effects, and the coefficient declines almost by nearly half, from 0.104 to 0.065. Column (5) substitutes census tract fixed effects with the more refined block group fixed effects, and the coefficient declines further. In Panel B, which uses presidential elections data, we

also observe a similar declining pattern in the coefficients. Most importantly, as I control for the most refined block group fixed effects, the correlation between owner-occupancy rate and presidential election turnout vanishes. This pattern is in line with the assumption that refined geographic fixed effects can absorb unobserved personal confounders associated with residential sorting. The residential sorting effect is weaker at a broader geographic level, thereby the broader fixed effects fail to capture the endogenous relationship between homeownership and voter turnout.

#### 4.2. Control function estimation results

The CF specification is given by Eq. (6). Compared to the fixed effects model, the CF approach allows the block group-level confounders to be time-variant. As discussed in Section 2.2, in this more flexible specification owner-occupancy rate should have limited impact on presidential election turnout. By assuming  $\beta_p = 0$ , I can identify  $\beta_m = 0$  which measures the impact of owner-occupancy rate on mayoral election turnout. To increase estimation precision, I pool the mayoral election records from all cities and election cycles together. City by election cycle fixed effects are included into the models to account for heterogeneity across city and time.

Table 8 presents the CF approach estimation results. Column (1) only includes owner-occupancy rate and neighborhood turnover rate as controls. The signs on both coefficients are consistent with previous findings from the fixed effects model. Column (2) adds presidential election turnout rate to control for unobserved confounders. The coefficient estimate on presidential election turnout rate is close to 0.45. Consistent with my expectation, the inclusion of presidential election turnout drives down the coefficient estimate on owner-occupancy rate. The coefficient estimate on the neighborhood turnover rate also decreases. Similar to the case of the owner-occupancy rate coefficient, after adding presidential election turnout as a control, the coefficient on the neighborhood turnover rate measures the differential impact of mobility on voter turnout in mayoral elections versus in presidential elections (scaled by  $\frac{\gamma_m}{\gamma_p}$ ). Therefore, the interpretation of this coefficient depends on the assumptions one is willing to make. Since the goal of including the neighborhood turnover rate is to control for mobility, I do not make extra assumptions to interpret this coefficient.

Column (3) adds the remaining census block group controls into the regression. The coefficients for these controls are not reported for the same reason as with the neighborhood turnover rate. As the result shows, conditional on mobility and other controls, the coefficient estimate on owner-occupancy rate  $\hat{\beta}_m$  is 0.049. This is close to the 0.038 estimate obtained from the fixed effects model (Table 4 Column (5)). The coefficient estimate on presidential election turnout remains stable. Results from the CF approach confirm the previous findings from the fixed effects model.

#### 4.3. Robustness check

To test the robustness of the results in both the fixed effects model and the CF approach, I conduct four sets of robustness checks. First, I drop census block groups located near major universities (U Penn, U Chicago and U of Washington) because those block groups contain a higher proportion of international students (non-U.S. citizens) which cannot be distinguished from the non-internal student population based on the census block group level data available. I also drop block groups located in non-residential areas such as harbors, factories or parks because demographic estimates from those areas may be less accurate. Those are just a handful of block groups and their elimination does not change any result. Second, to eliminate any concern that the results are driven by outliers or measurement error, I limit the sample to block



**Table 7**  
OLS estimation result comparison across different fixed effects models.

|                                 | (1)                | (2)                | (3)                | (4)                | (5)               |
|---------------------------------|--------------------|--------------------|--------------------|--------------------|-------------------|
| Panel A: Mayoral election       |                    |                    |                    |                    |                   |
| Owner-occupancy rate            | 0.079***<br>(0.01) | 0.083***<br>(0.01) | 0.104***<br>(0.01) | 0.065***<br>(0.01) | 0.038**<br>(0.02) |
| R-squared                       | 0.23               | 0.26               | 0.36               | 0.51               | 0.11              |
| Panel B: Presidential election  |                    |                    |                    |                    |                   |
| Owner-occupancy rate            | 0.136***<br>(0.01) | 0.130***<br>(0.01) | 0.101***<br>(0.01) | 0.081***<br>(0.01) | −0.034*<br>(0.02) |
| R-squared                       | 0.21               | 0.23               | 0.27               | 0.45               | 0.11              |
| Other slope variables           | Yes                | Yes                | Yes                | Yes                | Yes               |
| Election cycle fixed effect     | —                  | 3                  | 3                  | 3                  | 3                 |
| City fixed effect               | —                  | —                  | 3                  | —                  | —                 |
| Census tract fixed effect       | —                  | —                  | —                  | 1177               | —                 |
| Census block group fixed effect | —                  | —                  | —                  | —                  | 3369              |
| Observations                    | 10107              | 10107              | 10107              | 10107              | 10107             |

*Notes:* The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Panel A uses election panel data from mayoral elections. Panel B uses election panel data from presidential elections. Robust standard errors clustered at Census Blocks in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table 8**  
Control function estimation results using election panel data.

|                                                     | (1)                 | (2)                 | (3)                 |
|-----------------------------------------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.151***<br>(0.01)  | 0.120***<br>(0.00)  | 0.049***<br>(0.01)  |
| % Moved into current residence within past 12 month | −0.161***<br>(0.01) | −0.112***<br>(0.01) | −0.046***<br>(0.01) |
| Presidential election turnout                       |                     | 0.453***<br>(0.01)  | 0.451***<br>(0.01)  |
| R-squared                                           | 0.49                | 0.76                | 0.80                |
| City by election cycle fixed effects                | 9                   | 9                   | 9                   |
| Other slope controls                                | No                  | No                  | Yes                 |
| Observations                                        | 10107               | 10107               | 10107               |

*Notes:* The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census blocks in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

groups with turnout rate between 5% and 95% in all elections. Third, I run the specifications using the original full sample including census block groups with turnout larger than one.

In addition, to alleviate the concern that gentrification may invalidate the assumption that block group-level confounders are time-invariant in the fixed effects model, I conduct one more robustness check separately for the fixed effects model. I drop census block groups defined as been gentrified between 2000 and 2016 (about 10% of the original sample) by the *Governing* website<sup>22</sup> and find that all point estimates remain the same. I present all the robustness check tables in [Appendix A](#).

<sup>22</sup> The *Governing* website classifies gentrification at census tract level. A census tract is considered to be gentrified if the following two criteria are met. First, the tract median household income and median home value fall within the bottom 40th percentile of all tracts within a metro area in 2000. Second, the tract increases into the top third percentile for both inflation-adjusted median home values and percentage of adults with bachelors degrees in 2016. Details please refer to <http://www.governing.com/gov-data>.

## 5. Conclusion

Based on election data from three cities in the United States, this paper documents the fact that homeowners are more likely to vote in mayoral elections compared to renters living in the same community. There are two potential channels leading to this result. First, homeowners are more financially invested in the community by way of owning a house. Their financial stake in a community makes them more eager to participate in local political processes to promote policies that protect the property value. Renters, on the other hand, do not have similar stakes in a community and face fewer economic incentives to vote. Second, homeownership also creates frictions on household mobility. Therefore, homeowners receive longer utility flow from local public goods consumption. Controlling for the mobility channel, my results suggest that the financial incentive alone is the driver of homeowners' tendency to vote in mayoral elections. Identification in this paper is obtained by using fixed effects models and a control function approach. Both models use presidential elections as the counterfactual. Since national-level elections and polity initiatives are less relevant to property value and local public good provision, the difference in tendency to vote between homeowners and renters should be reduced in presidential elections.

The homevoter hypothesis literature shows home-voters and lease-voters vote differently in local political processes due to housing price capitalization effects ([Fischel, 2001](#); [Dehring et al., 2008](#); [Ahlfeldt and Maennig, 2015](#)). The findings in this paper further beg the question of whether uneven turnout between homeowners and renters in U.S. local elections causes biased policies favoring homeowners (e.g., using more strict zoning laws to keep housing value appreciation). In order to gain support from high propensity voters, local political candidates may be likely to design policies to please the high turnout group. This may result in policies that likely fail to maximize total social welfare. [McGregor and Spicer \(2016\)](#) finds that homeowners and renters have different opinions toward public housing projects. [Kahn \(2011\)](#) provides some evidence that homeowner cities are more likely to block new housing development based on data from California. Future work could marshal evidence as to whether uneven turnout between homeowners and renters affects the design of local policy or policy outcomes.

## Appendix A. Robustness check tables

### A1. Fixed effects model

**Table A.1**

Mayoral election fixed effects model results dropping non-residential block groups.

|                                                     | (1)                | (2)                 | (3)                | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.195***<br>(0.01) | 0.158***<br>(0.01)  | 0.078***<br>(0.01) | 0.065***<br>(0.01)  | 0.101***<br>(0.02)  | 0.052***<br>(0.02)  | 0.049***<br>(0.02)  | 0.040**<br>(0.02)   |
| % Moved into current residence within past 12 month |                    | −0.161***<br>(0.01) | −0.048**<br>(0.02) | −0.054***<br>(0.02) | −0.056***<br>(0.02) | −0.013<br>(0.02)    | −0.012<br>(0.02)    | −0.017<br>(0.02)    |
| % High school and some college among pop > 25       |                    |                     |                    | −0.116***<br>(0.02) | −0.106***<br>(0.02) | −0.086***<br>(0.02) | −0.085***<br>(0.02) | −0.101***<br>(0.02) |
| % College above among pop > 25                      |                    |                     |                    | 0.005<br>(0.02)     | 0.047*<br>(0.03)    | 0.087***<br>(0.02)  | 0.088***<br>(0.03)  | 0.043<br>(0.03)     |
| Household median income (Million \$)                |                    |                     |                    |                     | −0.828***<br>(0.12) | −0.787***<br>(0.12) | −0.821***<br>(0.13) | −0.853***<br>(0.13) |
| % Adults 30 to 60                                   |                    |                     |                    |                     |                     | 0.209***<br>(0.02)  | 0.206***<br>(0.02)  | 0.206***<br>(0.02)  |
| % Adults 60 above                                   |                    |                     |                    |                     |                     | 0.305***<br>(0.03)  | 0.304***<br>(0.03)  | 0.281***<br>(0.03)  |
| % Married-household                                 |                    |                     |                    |                     |                     |                     | 0.021<br>(0.02)     | 0.021<br>(0.02)     |
| % Hispanic                                          |                    |                     |                    |                     |                     |                     |                     | 0.001<br>(0.06)     |
| % White                                             |                    |                     |                    |                     |                     |                     |                     | 0.081<br>(0.06)     |
| % Black                                             |                    |                     |                    |                     |                     |                     |                     | −0.040<br>(0.06)    |
| % Asian                                             |                    |                     |                    |                     |                     |                     |                     | −0.028<br>(0.06)    |
| R-squared                                           | 0.24               | 0.25                | 0.06               | 0.07                | 0.07                | 0.10                | 0.10                | 0.11                |
| Election cycle fixed effects                        | 3                  | 3                   | 3                  | 3                   | 3                   | 3                   | 3                   | 3                   |
| City fixed effects                                  | 3                  | 3                   | —                  | —                   | —                   | —                   | —                   | —                   |
| Census block group fixed effects                    | —                  | —                   | 3324               | 3324                | 3324                | 3324                | 3324                | 3324                |
| Observations                                        | 9972               | 9972                | 9972               | 9972                | 9972                | 9972                | 9972                | 9972                |

*Notes:* The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.2**

Presidential election fixed effects model results dropping non-residential block groups.

|                                                     | (1)                | (2)                 | (3)              | (4)                | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|------------------|--------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.094***<br>(0.01) | 0.071***<br>(0.01)  | −0.017<br>(0.02) | −0.027<br>(0.02)   | 0.015<br>(0.02)     | −0.038*<br>(0.02)   | −0.026<br>(0.02)    | −0.035<br>(0.02)    |
| % Moved into current residence within past 12 month |                    | −0.103***<br>(0.02) | −0.038<br>(0.02) | −0.042*<br>(0.02)  | −0.045*<br>(0.02)   | 0.005<br>(0.02)     | 0.004<br>(0.02)     | 0.000<br>(0.02)     |
| % High school and some college among pop > 25       |                    |                     |                  | −0.053**<br>(0.03) | −0.042<br>(0.03)    | −0.021<br>(0.03)    | −0.027<br>(0.03)    | −0.034<br>(0.03)    |
| % College above among pop > 25                      |                    |                     |                  | 0.027<br>(0.03)    | 0.078***<br>(0.03)  | 0.121***<br>(0.03)  | 0.117***<br>(0.03)  | 0.085***<br>(0.03)  |
| Household median income (Million \$)                |                    |                     |                  |                    | −0.984***<br>(0.14) | −0.949***<br>(0.14) | −0.837***<br>(0.15) | −0.845***<br>(0.15) |
| % Adults 30 to 60                                   |                    |                     |                  |                    |                     | 0.246***<br>(0.03)  | 0.254***<br>(0.03)  | 0.259***<br>(0.03)  |
| % Adults 60 above                                   |                    |                     |                  |                    |                     | 0.337***<br>(0.03)  | 0.340***<br>(0.03)  | 0.328***<br>(0.03)  |
| % Married-household                                 |                    |                     |                  |                    |                     |                     | −0.066***<br>(0.02) | −0.072***<br>(0.02) |
| % Hispanic                                          |                    |                     |                  |                    |                     |                     |                     | 0.059<br>(0.07)     |
| % White                                             |                    |                     |                  |                    |                     |                     |                     | 0.101<br>(0.07)     |
| % Black                                             |                    |                     |                  |                    |                     |                     |                     | −0.024              |

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**Table A.2** (continued)

|                                  | (1)  | (2)  | (3)  | (4)  | (5)  | (6)  | (7)  | (8)                       |
|----------------------------------|------|------|------|------|------|------|------|---------------------------|
| % Asian                          |      |      |      |      |      |      |      | (0.07)<br>0.064<br>(0.07) |
| R-squared                        | 0.10 | 0.10 | 0.07 | 0.07 | 0.08 | 0.10 | 0.10 | 0.11                      |
| Election cycle fixed effects     | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3                         |
| City fixed effects               | 3    | 3    | —    | —    | —    | —    | —    | —                         |
| Census block group fixed effects | —    | —    | 3324 | 3324 | 3324 | 3324 | 3324 | 3324                      |
| Observations                     | 9972 | 9972 | 9972 | 9972 | 9972 | 9972 | 9972 | 9972                      |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.3**

Mayoral election fixed effects model results limited to blocks with turnout rate between 5% and 95%.

|                                                     | (1)                | (2)                 | (3)                | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.188***<br>(0.01) | 0.154***<br>(0.01)  | 0.079***<br>(0.01) | 0.067***<br>(0.02)  | 0.100***<br>(0.02)  | 0.055***<br>(0.02)  | 0.051***<br>(0.02)  | 0.043**<br>(0.02)   |
| % Moved into current residence within past 12 month |                    | −0.149***<br>(0.01) | −0.042**<br>(0.02) | −0.048**<br>(0.02)  | −0.050**<br>(0.02)  | −0.006<br>(0.02)    | −0.006<br>(0.02)    | −0.011<br>(0.02)    |
| % High school and some college among pop > 25       |                    |                     |                    | −0.115***<br>(0.02) | −0.106***<br>(0.02) | −0.084***<br>(0.02) | −0.082***<br>(0.02) | −0.097***<br>(0.02) |
| % College above among pop > 25                      |                    |                     |                    | 0.005<br>(0.03)     | 0.043*<br>(0.03)    | 0.084***<br>(0.03)  | 0.086***<br>(0.03)  | 0.043<br>(0.03)     |
| Household median income (Million \$)                |                    |                     |                    |                     | −0.779***<br>(0.12) | −0.749***<br>(0.12) | −0.779***<br>(0.13) | −0.808***<br>(0.13) |
| % Adults 30 to 60                                   |                    |                     |                    |                     |                     | 0.207***<br>(0.02)  | 0.205***<br>(0.02)  | 0.203***<br>(0.02)  |
| % Adults 60 above                                   |                    |                     |                    |                     |                     | 0.301***<br>(0.03)  | 0.300***<br>(0.03)  | 0.277***<br>(0.03)  |
| % Married-household                                 |                    |                     |                    |                     |                     |                     | 0.019<br>(0.02)     | 0.020<br>(0.02)     |
| raceSH_hispanic                                     |                    |                     |                    |                     |                     |                     |                     | −0.011<br>(0.06)    |
| raceSH_white                                        |                    |                     |                    |                     |                     |                     |                     | 0.063<br>(0.06)     |
| raceSH_black                                        |                    |                     |                    |                     |                     |                     |                     | −0.050<br>(0.06)    |
| raceSH_asian                                        |                    |                     |                    |                     |                     |                     |                     | −0.050<br>(0.06)    |
| R-squared                                           | 0.23               | 0.24                | 0.06               | 0.07                | 0.08                | 0.10                | 0.10                | 0.11                |
| Election cycle fixed effect                         | 3                  | 3                   | 3                  | 3                   | 3                   | 3                   | 3                   | 3                   |
| City fixed effect                                   | 3                  | 3                   | —                  | —                   | —                   | —                   | —                   | —                   |
| Census block group fixed effect                     | —                  | —                   | 3144               | 3144                | 3144                | 3144                | 3144                | 3144                |
| Observations                                        | 9432               | 9432                | 9432               | 9432                | 9432                | 9432                | 9432                | 9432                |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.4**

Presidential election fixed effects model results limited to blocks with turnout rate between 5% and 95%.

|                                                     | (1)                | (2)                 | (3)               | (4)                | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|-------------------|--------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.085***<br>(0.01) | 0.066***<br>(0.01)  | −0.017<br>(0.02)  | −0.027<br>(0.02)   | 0.016<br>(0.02)     | −0.034*<br>(0.02)   | −0.023<br>(0.02)    | −0.032<br>(0.02)    |
| % Moved into current residence within past 12 month |                    | −0.080***<br>(0.02) | −0.042*<br>(0.02) | −0.047**<br>(0.02) | −0.048**<br>(0.02)  | 0.002<br>(0.02)     | 0.001<br>(0.02)     | −0.003<br>(0.02)    |
| % High school and some college among pop > 25       |                    |                     |                   | −0.059**<br>(0.03) | −0.048*<br>(0.03)   | −0.025<br>(0.03)    | −0.031<br>(0.03)    | −0.037<br>(0.03)    |
| % College above among pop > 25                      |                    |                     |                   | 0.025<br>(0.03)    | 0.073**<br>(0.03)   | 0.118***<br>(0.03)  | 0.114***<br>(0.03)  | 0.086***<br>(0.03)  |
| Household median income (Million \$)                |                    |                     |                   |                    | −0.984***<br>(0.14) | −0.973***<br>(0.14) | −0.881***<br>(0.14) | −0.883***<br>(0.14) |

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**Table A.4** (continued)

|                                 | (1)  | (2)  | (3)  | (4)  | (5)  | (6)                | (7)                 | (8)                 |
|---------------------------------|------|------|------|------|------|--------------------|---------------------|---------------------|
| % Adults 30 to 60               |      |      |      |      |      | 0.256***<br>(0.03) | 0.263***<br>(0.03)  | 0.267***<br>(0.03)  |
| % Adults 60 above               |      |      |      |      |      | 0.326***<br>(0.03) | 0.328***<br>(0.03)  | 0.318***<br>(0.03)  |
| % Married-household             |      |      |      |      |      |                    | −0.058***<br>(0.02) | −0.063***<br>(0.02) |
| raceSH_hispanic                 |      |      |      |      |      |                    |                     | 0.062<br>(0.07)     |
| raceSH_white                    |      |      |      |      |      |                    |                     | 0.093<br>(0.06)     |
| raceSH_black                    |      |      |      |      |      |                    |                     | −0.021<br>(0.06)    |
| raceSH_asian                    |      |      |      |      |      |                    |                     | 0.055<br>(0.07)     |
| R-squared                       | 0.09 | 0.10 | 0.07 | 0.07 | 0.08 | 0.11               | 0.11                | 0.11                |
| Election cycle fixed effect     | 3    | 3    | 3    | 3    | 3    | 3                  | 3                   | 3                   |
| City fixed effect               | 3    | 3    | —    | —    | —    | —                  | —                   | —                   |
| Census block group fixed effect | —    | —    | 3144 | 3144 | 3144 | 3144               | 3144                | 3144                |
| Observations                    | 9432 | 9432 | 9432 | 9432 | 9432 | 9432               | 9432                | 9432                |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.5**

Mayoral election fixed effects model results including blocks with turnout rate larger than one.

|                                                     | (1)                | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.213***<br>(0.01) | 0.176***<br>(0.01)  | 0.082***<br>(0.01)  | 0.069***<br>(0.02)  | 0.100***<br>(0.02)  | 0.048***<br>(0.02)  | 0.048***<br>(0.02)  | 0.040**<br>(0.02)   |
| % Moved into current residence within past 12 month |                    | −0.160***<br>(0.01) | −0.053***<br>(0.02) | −0.058***<br>(0.02) | −0.059***<br>(0.02) | −0.009<br>(0.02)    | −0.009<br>(0.02)    | −0.014<br>(0.02)    |
| % High school and some college among pop > 25       |                    |                     |                     | −0.131***<br>(0.02) | −0.123***<br>(0.02) | −0.103***<br>(0.02) | −0.102***<br>(0.02) | −0.119***<br>(0.02) |
| % College above among pop > 25                      |                    |                     |                     | −0.006<br>(0.02)    | 0.029<br>(0.03)     | 0.069***<br>(0.02)  | 0.069***<br>(0.02)  | 0.028<br>(0.03)     |
| Household median income (Million \$)                |                    |                     |                     |                     | −0.693***<br>(0.12) | −0.669***<br>(0.12) | −0.676***<br>(0.12) | −0.703***<br>(0.12) |
| % Adults 30 to 60                                   |                    |                     |                     |                     |                     | 0.239***<br>(0.02)  | 0.238***<br>(0.02)  | 0.239***<br>(0.02)  |
| % Adults 60 above                                   |                    |                     |                     |                     |                     | 0.329***<br>(0.03)  | 0.329***<br>(0.03)  | 0.309***<br>(0.03)  |
| % Married-household                                 |                    |                     |                     |                     |                     |                     | 0.004<br>(0.02)     | 0.005<br>(0.02)     |
| raceSH_hispanic                                     |                    |                     |                     |                     |                     |                     |                     | −0.006<br>(0.06)    |
| raceSH_white                                        |                    |                     |                     |                     |                     |                     |                     | 0.070<br>(0.06)     |
| raceSH_black                                        |                    |                     |                     |                     |                     |                     |                     | −0.041<br>(0.06)    |
| raceSH_asian                                        |                    |                     |                     |                     |                     |                     |                     | −0.047<br>(0.06)    |
| R-squared                                           | 0.25               | 0.26                | 0.06                | 0.06                | 0.07                | 0.10                | 0.10                | 0.10                |
| Election cycle fixed effects                        | 3                  | 3                   | 3                   | 3                   | 3                   | 3                   | 3                   | 3                   |
| City fixed effects                                  | 3                  | 3                   | —                   | —                   | —                   | —                   | —                   | —                   |
| Census block group fixed effects                    | —                  | —                   | 3912                | 3912                | 3912                | 3912                | 3912                | 3912                |
| Observations                                        | 11658              | 11658               | 11658               | 11658               | 11658               | 11658               | 11658               | 11658               |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.



**Table A.6**

Presidential election fixed effects model results including blocks with turnout rate larger than one.

|                                                     | (1)                | (2)                 | (3)              | (4)               | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|------------------|-------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.104***<br>(0.01) | 0.077***<br>(0.01)  | −0.002<br>(0.02) | −0.013<br>(0.02)  | 0.026<br>(0.02)     | −0.026<br>(0.02)    | −0.014<br>(0.02)    | −0.023<br>(0.02)    |
| % Moved into current residence within past 12 month |                    | −0.113***<br>(0.02) | −0.028<br>(0.02) | −0.032<br>(0.02)  | −0.034<br>(0.02)    | 0.017<br>(0.02)     | 0.016<br>(0.02)     | 0.011<br>(0.02)     |
| % High school and some college among pop > 25       |                    |                     |                  | −0.048*<br>(0.03) | −0.038<br>(0.03)    | −0.014<br>(0.03)    | −0.020<br>(0.03)    | −0.027<br>(0.03)    |
| % College above among pop > 25                      |                    |                     |                  | 0.040<br>(0.03)   | 0.085***<br>(0.03)  | 0.130***<br>(0.03)  | 0.127***<br>(0.03)  | 0.098***<br>(0.03)  |
| Household median income (Million \$)                |                    |                     |                  |                   | −0.910***<br>(0.14) | −0.896***<br>(0.14) | −0.791***<br>(0.15) | −0.789***<br>(0.15) |
| % Adults 30 to 60                                   |                    |                     |                  |                   |                     | 0.253***<br>(0.03)  | 0.261***<br>(0.03)  | 0.267***<br>(0.03)  |
| % Adults 60 above                                   |                    |                     |                  |                   |                     | 0.334***<br>(0.03)  | 0.338***<br>(0.03)  | 0.328***<br>(0.03)  |
| % Married-household                                 |                    |                     |                  |                   |                     |                     | −0.066***<br>(0.02) | −0.073***<br>(0.02) |
| raceSH_hispanic                                     |                    |                     |                  |                   |                     |                     |                     | 0.066<br>(0.06)     |
| raceSH_white                                        |                    |                     |                  |                   |                     |                     |                     | 0.101<br>(0.06)     |
| raceSH_black                                        |                    |                     |                  |                   |                     |                     |                     | −0.014<br>(0.06)    |
| raceSH_asian                                        |                    |                     |                  |                   |                     |                     |                     | 0.076<br>(0.07)     |
| R-squared                                           | 0.10               | 0.11                | 0.08             | 0.08              | 0.09                | 0.11                | 0.11                | 0.12                |
| Election cycle fixed effects                        | 3                  | 3                   | 3                | 3                 | 3                   | 3                   | 3                   | 3                   |
| City fixed effects                                  | 3                  | 3                   | —                | —                 | —                   | —                   | —                   | —                   |
| Census block group fixed effects                    | —                  | —                   | 3889             | 3889              | 3889                | 3889                | 3889                | 3889                |
| Observations                                        | 10820              | 10820               | 10820            | 10820             | 10820               | 10820               | 10820               | 10820               |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.7**

Mayoral election fixed effects model results dropping gentrified census blocks.

|                                                     | (1)                | (2)                 | (3)                | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.200***<br>(0.01) | 0.163***<br>(0.01)  | 0.085***<br>(0.02) | 0.062***<br>(0.02)  | 0.091***<br>(0.02)  | 0.048***<br>(0.02)  | 0.045**<br>(0.02)   | 0.032*<br>(0.02)    |
| % Moved into current residence within past 12 month |                    | −0.158***<br>(0.01) | −0.043**<br>(0.02) | −0.050**<br>(0.02)  | −0.052***<br>(0.02) | −0.010<br>(0.02)    | −0.009<br>(0.02)    | −0.016<br>(0.02)    |
| % High school and some college among pop > 25       |                    |                     |                    | −0.126***<br>(0.02) | −0.119***<br>(0.02) | −0.098***<br>(0.02) | −0.096***<br>(0.02) | −0.114***<br>(0.02) |
| % College above among pop > 25                      |                    |                     |                    | 0.065**<br>(0.03)   | 0.097***<br>(0.03)  | 0.132***<br>(0.03)  | 0.133***<br>(0.03)  | 0.081***<br>(0.03)  |
| Household median income (Million \$)                |                    |                     |                    |                     | −0.652***<br>(0.12) | −0.652***<br>(0.12) | −0.681***<br>(0.12) | −0.699***<br>(0.12) |
| % Adults 30 to 60                                   |                    |                     |                    |                     |                     | 0.210***<br>(0.02)  | 0.208***<br>(0.02)  | 0.209***<br>(0.02)  |
| % Adults 60 above                                   |                    |                     |                    |                     |                     | 0.273***<br>(0.03)  | 0.272***<br>(0.03)  | 0.247***<br>(0.03)  |
| % Married-household                                 |                    |                     |                    |                     |                     |                     | 0.017<br>(0.02)     | 0.015<br>(0.02)     |
| raceSH_hispanic                                     |                    |                     |                    |                     |                     |                     |                     | −0.039<br>(0.06)    |
| raceSH_white                                        |                    |                     |                    |                     |                     |                     |                     | 0.055<br>(0.06)     |
| raceSH_black                                        |                    |                     |                    |                     |                     |                     |                     | −0.106*<br>(0.06)   |
| raceSH_asian                                        |                    |                     |                    |                     |                     |                     |                     | −0.027<br>(0.07)    |
| R-squared                                           | 0.25               | 0.26                | 0.04               | 0.06                | 0.06                | 0.09                | 0.09                | 0.10                |
| Election cycle fixed effects                        | 3                  | 3                   | 3                  | 3                   | 3                   | 3                   | 3                   | 3                   |
| City fixed effects                                  | 3                  | 3                   | —                  | —                   | —                   | —                   | —                   | —                   |

(continued on next page)

**Table A.7** (continued)

|                                  | (1)  | (2)  | (3)  | (4)  | (5)  | (6)  | (7)  | (8)  |
|----------------------------------|------|------|------|------|------|------|------|------|
| Census block group fixed effects | —    | —    | 3005 | 3005 | 3005 | 3005 | 3005 | 3005 |
| Observations                     | 8967 | 8967 | 8967 | 8967 | 8967 | 8967 | 8967 | 8967 |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.8**

Presidential election fixed effects model results dropping gentrified census blocks.

|                                                     | (1)                | (2)                 | (3)               | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|-----------------------------------------------------|--------------------|---------------------|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.097***<br>(0.01) | 0.073***<br>(0.01)  | −0.012<br>(0.02)  | −0.025<br>(0.02)    | 0.016<br>(0.02)     | −0.032<br>(0.02)    | −0.019<br>(0.02)    | −0.028<br>(0.02)    |
| % Moved into current residence within past 12 month |                    | −0.101***<br>(0.02) | −0.041*<br>(0.02) | −0.046*<br>(0.02)   | −0.048**<br>(0.02)  | 0.001<br>(0.02)     | −0.000<br>(0.02)    | −0.004<br>(0.02)    |
| % High school and some college among pop > 25       |                    |                     |                   | −0.071***<br>(0.03) | −0.060**<br>(0.03)  | −0.038<br>(0.03)    | −0.044<br>(0.03)    | −0.052*<br>(0.03)   |
| % College above among pop > 25                      |                    |                     |                   | 0.035<br>(0.03)     | 0.082***<br>(0.03)  | 0.121***<br>(0.03)  | 0.117***<br>(0.03)  | 0.090***<br>(0.03)  |
| Household median income (Million \$)                |                    |                     |                   |                     | −0.932***<br>(0.14) | −0.937***<br>(0.14) | −0.821***<br>(0.15) | −0.823***<br>(0.15) |
| % Adults 30 to 60                                   |                    |                     |                   |                     |                     | 0.253***<br>(0.03)  | 0.261***<br>(0.03)  | 0.264***<br>(0.03)  |
| % Adults 60 above                                   |                    |                     |                   |                     |                     | 0.304***<br>(0.03)  | 0.307***<br>(0.03)  | 0.295***<br>(0.03)  |
| % Married-household                                 |                    |                     |                   |                     |                     |                     | −0.070***<br>(0.02) | −0.074***<br>(0.02) |
| raceSH_hispanic                                     |                    |                     |                   |                     |                     |                     |                     | 0.065<br>(0.07)     |
| raceSH_white                                        |                    |                     |                   |                     |                     |                     |                     | 0.110<br>(0.07)     |
| raceSH_black                                        |                    |                     |                   |                     |                     |                     |                     | 0.007<br>(0.07)     |
| raceSH_asian                                        |                    |                     |                   |                     |                     |                     |                     | 0.090<br>(0.07)     |
| R-squared                                           | 0.10               | 0.10                | 0.07              | 0.07                | 0.08                | 0.11                | 0.11                | 0.11                |
| Election cycle fixed effects                        | 3                  | 3                   | 3                 | 3                   | 3                   | 3                   | 3                   | 3                   |
| City fixed effects                                  | 3                  | 3                   | —                 | —                   | —                   | —                   | —                   | —                   |
| Census block group fixed effects                    | —                  | —                   | 3007              | 3007                | 3007                | 3007                | 3007                | 3007                |
| Observations                                        | 8848               | 8848                | 8848              | 8848                | 8848                | 8848                | 8848                | 8848                |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census block groups in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

## A2. Control function approach

**Table A.9**

Control function estimation results dropping non-residential block groups.

|                                                     | (1)                 | (2)                 | (3)                 |
|-----------------------------------------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                | 0.148***<br>(0.01)  | 0.118***<br>(0.00)  | 0.050***<br>(0.01)  |
| % Moved into current residence within past 12 month | −0.140***<br>(0.01) | −0.106***<br>(0.01) | −0.041***<br>(0.01) |
| Presidential election turnout                       |                     | 0.450***<br>(0.01)  | 0.452***<br>(0.01)  |
| R-squared                                           | 0.49                | 0.75                | 0.79                |
| City by election cycle fixed effect                 | 9                   | 9                   | 9                   |
| Other slope controls                                | No                  | No                  | Yes                 |
| Observations                                        | 9324                | 9324                | 9324                |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census blocks in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.10**

Control function estimation results limited to block groups with turnout rate between 5% and 95%.

|                                                        | (1)                 | (2)                 | (3)                 |
|--------------------------------------------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                   | 0.147***<br>(0.01)  | 0.118***<br>(0.00)  | 0.050***<br>(0.01)  |
| % Moved into current residence within<br>past 12 month | −0.145***<br>(0.01) | −0.108***<br>(0.01) | −0.042***<br>(0.01) |
| Presidential election turnout                          |                     | 0.451***<br>(0.01)  | 0.452***<br>(0.01)  |
| R-squared                                              | 0.49                | 0.75                | 0.79                |
| City by election cycle fixed effects                   | 9                   | 9                   | 9                   |
| Other slope controls                                   | No                  | No                  | Yes                 |
| Observations                                           | 9432                | 9432                | 9432                |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census blocks in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

**Table A.11**

Control function estimation results including block groups with turnout rate larger than one.

|                                                        | (1)                 | (2)                 | (3)                 |
|--------------------------------------------------------|---------------------|---------------------|---------------------|
| Owner-occupancy rate                                   | 0.172***<br>(0.01)  | 0.137***<br>(0.00)  | 0.056***<br>(0.01)  |
| % Moved into current residence within<br>past 12 month | −0.167***<br>(0.01) | −0.113***<br>(0.01) | −0.059***<br>(0.01) |
| Presidential election turnout                          |                     | 0.430***<br>(0.01)  | 0.425***<br>(0.01)  |
| R-squared                                              | 0.43                | 0.78                | 0.81                |
| City by election cycle fixed effects                   | 9                   | 9                   | 9                   |
| Other slope controls                                   | No                  | No                  | Yes                 |
| Observations                                           | 11757               | 11757               | 11757               |

Notes: The election panel data is from Philadelphia, Chicago and Seattle between 2003 and 2013. The unit of observation is census block groups. Robust standard errors clustered at census blocks in parenthesis. \*\*\* Significant at 1%, \*\* significant at 5%, \* significant at 10%.

## Appendix B. Data sources

### B1. Election data

Philadelphia election data is retrieved from Philadelphia City Commissioners website: <http://www.philadelphiavotes.com/en/resources-a-data/ballot-box-app>. Chicago election data is retrieved from Chicago Board of Election Commissioners website: <http://www.chicagoelections.com/en/election3.asp>. Seattle election data is retrieved from King County's county website: <http://www.kingcounty.gov/depts/elections/elections/past-elections.aspx>.

In Chicago, the precinct definition changed between year 2011 and 2012. To append three rounds Chicago election data together I manually build a correspondent between old and new Chicago precinct using GIS software. The methodology is the same as the one I use to build correspondence between 2000 census block groups and 2010 census block groups. Detail of the GIS work is laid out in [Appendix C](#).

### B2. Census data

Most of the census block groups level demographic data is not available on the American Fact Finder (<https://factfinder.census.gov/faces/nav/jsf/pages/index.xhtml>). I retrieve the data from ACS Summary file on Census server and clean them following the manual book provided by Census. Detail about the Summary file and retrieving guidance is available at <http://www.census.gov/programs-surveys/acs/data/tools/summary-file-retrieval-tool.html>.

The Voting Age Population by Citizenship and Race (CVAP) data is a separate product by Census. It's available at [https://www.census.gov/rdo/data/voting\\_age\\_population\\_by\\_citizenship\\_and\\_race\\_cvap.html](https://www.census.gov/rdo/data/voting_age_population_by_citizenship_and_race_cvap.html).

## Appendix C. Census block groups correspondence file building procedure

For each of the three cities, I use GIS software to build the correspondence between 2000 census block groups and 2010 census block groups in following steps:

(1) Obtain the 2000 and 2010 census block groups shapefiles from Census website <https://www.census.gov/geo/maps-data/data/tiger-cart-boundary.html>.

(2) Overlap the 2000 and 2010 shapefiles together in the GIS software.

(3) For each 2000 census block groups, the GIS software calculate its overlapping with all 2010 census block groups. Then based on the calculation the GIS software will assign a number to indicate what percentage of a 2000 census block groups belongs to a 2010 census block groups. This generates a correspondence file for the given city.

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